



## **USL Assignment 2**

### **Implement the principal component analysis algorithm**

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#### **PCA THEORY**

##### **1. Introduction to Principal Component Analysis**

Principal Component Analysis (PCA) is a statistical dimensionality reduction technique used to transform a dataset containing multiple possibly correlated variables into a new coordinate system of uncorrelated variables known as principal components.

The transformation is performed in such a way that:

- the first principal component captures the maximum variance in the dataset
- the second principal component captures the next highest variance



- subsequent components capture remaining variance in decreasing order

Thus, PCA allows reduction of feature space while preserving maximum information.

## 2. Mathematical Concept Behind PCA

Given a dataset with multiple variables:

1. Data is first standardized so that each feature has mean 0 and variance 1
2. A covariance matrix is computed to measure relationships between features
3. Eigenvalues and eigenvectors of the covariance matrix are calculated
  - Eigenvectors → directions of principal components
  - Eigenvalues → amount of variance captured

The eigenvectors corresponding to the largest eigenvalues form the reduced feature space.



### 3. Steps of PCA Algorithm

1. Standardize the dataset
2. Compute covariance matrix
3. Compute eigenvalues and eigenvectors
4. Sort eigenvalues in descending order
5. Select top k eigenvectors
6. Transform original dataset into reduced dimensional space

### 4. Example of PCA

Consider a dataset containing two highly correlated health measurements:

- cholesterol level
- LDL lipid measurement

Both represent related cardiovascular risk.

Instead of using both variables separately, PCA combines them into one principal component representing overall lipid health.

This reduces redundancy while preserving important information for medical prediction.



## 5. PCA Applied to Cleveland Heart Disease Dataset

The Cleveland Heart Disease dataset consists of multiple clinical measurements describing patient cardiovascular health, including blood pressure, cholesterol, ECG readings, and exercise indicators.

Many of these medical parameters are correlated and jointly influence heart disease diagnosis. Therefore PCA was applied to:

- identify dominant combined health factors
- reduce redundancy among correlated measurements
- transform dataset into principal component space

## 6. Inferences from PCA Output

After applying PCA to the Cleveland Heart Disease dataset, the explained variance ratio showed that the first principal component captured the largest proportion of total variance, followed by progressively smaller contributions from subsequent components. The variance distribution indicates that no single medical attribute dominates the dataset; instead, multiple physiological measurements jointly contribute to cardiovascular risk assessment. The cumulative explained variance plot shows that several principal components are required to retain most of the dataset information, confirming that heart disease diagnosis depends on a combination of clinical factors. PCA successfully reduced dimensionality while preserving the essential structure of the medical data.



## 7. References

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>

<https://archive.ics.uci.edu/>

Jolliffe, I.T., Principal Component Analysis, Springer.